

Apollo Diagnostics - Laboratory Investigation Report

123, Nungambakkam High Road, Chennai - 600034 | NABL Accredited | Tel: 044-2827-5000

Patient Name:	Dinesh Pandey	Report ID:	APL-META-48-DIN-2026
Age / Gender:	48 Yrs / Male	Category:	Metabolic Syndrome
Weight / Height:	97 kg / 175 cm	BMI:	31.7 kg/m2
Ref. Doctor:	Dr. Sunita Reddy	Report Date:	25-Jan-2026

1. COMPLETE BLOOD COUNT [Circulatory System]

Red Cell Indices

Test	Result	Unit	Reference Range	Flag
Haemoglobin (Hb)	14.2	g/dL	13.0 - 17.5	Normal
Packed Cell Volume (PCV)	43.0	%	40 - 52	Normal
MCV	90.0	fL	83 - 101	Normal
MCH	29.0	pg	27 - 32	Normal
RDW-CV	13.0	%	11.6 - 14.0	Normal

Haematology Comment: Haemoglobin 14.2 g/dL is within normal range. Red cell indices are unremarkable.

White Cells & Platelets

Test	Result	Unit	Reference Range	Flag
Total WBC	7000	cells/cmm	4500 - 11000	Normal
Neutrophils	62	%	40 - 75	Normal
Lymphocytes	28	%	20 - 45	Normal
Platelet Count	2.8	lakh/cmm	1.5 - 4.0	Normal

Comment: WBC and platelet counts within reference limits. No significant leucocytosis, leucopenia or thrombocytopenia.

2. IRON STUDIES [Haematopoietic System]

Test	Result	Unit	Reference Range	Flag
Serum Iron	110	ug/dL	60 - 170	Normal
TIBC	310	ug/dL	250 - 370	Normal
Transferrin Saturation	35.0	%	20 - 50	Normal
Serum Ferritin	80.0	ng/mL	30 - 300	Normal

Iron Studies Comment: Iron stores and transport markers are within normal limits. No iron deficiency or iron overload.

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3. HEPATIC FUNCTION TEST [Hepatobiliary System]

Liver Enzymes

Test	Result	Unit	Reference Range	Flag
SGPT / ALT	73	U/L	7 - 56	HIGH
SGOT / AST	72	U/L	10 - 40	HIGH
Alkaline Phosphatase (ALP)	70	U/L	44 - 147	Normal
Total Bilirubin	0.6	mg/dL	0.2 - 1.2	Normal
Albumin	4.2	g/dL	3.5 - 5.0	Normal

Hepatologist Note: ALT 73 U/L and AST 72 U/L are elevated indicating hepatocellular injury. Likely causes include NAFLD, alcohol use, or viral hepatitis. Liver ultrasound and viral hepatitis screen recommended.

4. RENAL FUNCTION TEST [Urinary System]

Test	Result	Unit	Reference Range	Flag
Serum Creatinine	1.0	mg/dL	0.7 - 1.3	Normal
Blood Urea Nitrogen	14	mg/dL	7 - 20	Normal
eGFR (CKD-EPI)	64	mL/min/1.73m2	60 - 120	Normal
Serum Uric Acid	4.8	mg/dL	2.4 - 7.0	Normal
Sodium (Na+)	139	mEq/L	135 - 145	Normal
Potassium (K+)	4.0	mEq/L	3.5 - 5.1	Normal

Nephrologist Note: eGFR 64 mL/min/1.73m2 - glomerular filtration well preserved. Renal function normal.

5. LIPID PROFILE [Cardiovascular Risk]

Test	Result	Unit	Reference Range	Flag
Total Cholesterol	289	mg/dL	0 - 200	HIGH
HDL Cholesterol	36	mg/dL	40 - 999	LOW
LDL Cholesterol	165	mg/dL	0 - 100	HIGH
Triglycerides	308	mg/dL	0 - 150	HIGH

Cardiologist Note: LDL 165 mg/dL (target <100 mg/dL) and Total Cholesterol 289 mg/dL are elevated. Triglycerides 308 mg/dL. Increased cardiovascular risk. RECOMMENDATION: Mediterranean diet, aerobic exercise, statin therapy consideration if >10-year CV risk >7.5%.

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6. THYROID FUNCTION TEST [Endocrine System]

Test	Result	Unit	Reference Range	Flag
TSH	4.7	uIU/mL	0.4 - 4.0	HIGH
Free T3	3.2	pg/mL	2.3 - 4.2	Normal
Free T4	1.1	ng/dL	0.8 - 1.8	Normal

Endocrinologist Comment: TSH: 4.7 uIU/mL is elevated - consistent with hypothyroidism. FT4 1.1 ng/dL is borderline (subclinical hypothyroidism). Thyroid antibodies (Anti-TPO) and ultrasound recommended. Consider levothyroxine therapy.

7. DIABETES SCREENING [Endocrine - Pancreas]

Test	Result	Unit	Reference Range	Flag
Fasting Blood Glucose	124	mg/dL	70 - 100	HIGH
Post-Prandial Glucose (2hr)	214	mg/dL	0 - 140	HIGH
HbA1c	8.4	%	0 - 5.7	HIGH
HOMA-IR	6.2		0 - 2.5	HIGH

Endocrinology - CRITICAL FINDING: HbA1c 8.4% and FBG 124 mg/dL confirm DIABETES MELLITUS TYPE 2. PPG 214 mg/dL. Immediate antidiabetic therapy (Metformin 500-1000 mg) and dietary modification required. Quarterly HbA1c monitoring and annual fundoscopy/nephropathy screening advised.

8. VITAMINS & BONE [Nutritional / Skeletal]

Test	Result	Unit	Reference Range	Flag
Vitamin D (25-OH)	10.1	ng/mL	30 - 100	LOW
Vitamin B12	450	pg/mL	211 - 911	Normal
Folate (Serum)	12.0	ng/mL	5.4 - 999	Normal
Calcium (Total)	9.4	mg/dL	8.5 - 10.5	Normal
PTH	32	pg/mL	15 - 65	Normal

Nutritionist/Endocrinologist Comment: Vitamin D: 10 ng/mL - deficient (10-20). PTH at 32 pg/mL is within range. Supplementation: 60,000 IU cholecalciferol weekly x 8 weeks, then 1000-2000 IU/day. Recheck in 3 months.

9. INFLAMMATORY MARKERS [Immune System]

Test	Result	Unit	Reference Range	Flag
ESR	38	mm/hr	0 - 15	HIGH
CRP	11.5	mg/L	0 - 5	HIGH
hs-CRP	3.8	mg/L	0 - 1.0	HIGH

Immunology Comment: CRP 11.5 mg/L and ESR 38 mm/hr are elevated. hs-CRP 3.8 mg/L - high CV risk (>3). Inflammatory markers elevated - correlate with clinical symptoms and autoimmune screen.

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CLINICAL SUMMARY & RECOMMENDATIONS

[CRITICAL]	Diabetes Mellitus Type 2 confirmed: HbA1c=8.4%, FBG=124 mg/dL. Antidiabetic therapy required.
[ELEVATED]	LDL=165 mg/dL (HIGH). Total Cholesterol=289. Dyslipidaemia - dietary + pharmacological management.
[ABNORMAL]	Hypothyroidism: TSH=4.7 uIU/mL. Levothyroxine therapy evaluation required.
[ELEVATED]	Elevated liver enzymes: ALT=73, AST=72. Liver disease workup required.
[DEFICIENT]	Vitamin D=10 ng/mL (LOW). Supplementation required.
[HIGH]	CRP=11.5 mg/L, hs-CRP=3.8 mg/L - systemic inflammation present.
[ACTION]	Next appointment: Dr. Sunita Reddy in 4-8 weeks for follow-up and review of supplementation response.

Report reviewed by: Dr. Anitha Krishnamurthy, MD (Pathology) | Verified by Dr. Sunita Reddy

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